

ARYAN RAUT

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in Aryan Raut

AryanRaut14

Summary

Computer Science undergraduate specializing in AI and LLM Engineering. Experienced in building agentic LLM orchestration systems, designing structured prompt-engineering pipelines, and architecting resilient, multi-provider inference systems with local and cloud model fallback.

Education

Atharva College of Engineering

B.E. in Computer Science (GPA: 8.83/10)

2022 – 2026

Mumbai, Maharashtra

Projects

Culer – Agentic LLM Assistant for FC Barcelona

LangGraph.js, Next.js, TypeScript, Ollama, Google Gemini API, Groq API

- Architected a LangGraph.js-based agentic pipeline with a dedicated router node that uses structured JSON-output prompting to classify query intent and conditionally trigger live web search, improving reliability over implicit tool-calling on smaller local models.
- Engineered a resilient multi-provider fallback strategy (local Ollama primary, Google Gemini and Groq cloud fallback) that dynamically shifts inference during offline or rate-limited states to eliminate downtime, with LangGraph checkpointing to restore multi-turn context per browser-generated conversation thread.
- Designed a Next.js/TypeScript conversational UI with a streamed chat interface, delivering privacy-conscious, zero-tracking query responses.

Dual-LLM Sentiment Analysis & Summarization Engine

Python, Streamlit, Scikit-Learn (LinearSVC), Gemini API, Groq, NLTK

- Trained an end-to-end sentiment classification pipeline on 1.6M+ tweets, incorporating negation-aware text preprocessing and trigram TF-IDF vectorization to achieve 78.16% accuracy using a Calibrated LinearSVC model.
- Designed prompt-engineered summarization chains across Google Gemini 3.5 Flash and Groq (LLaMA 3.3 70B) with automatic failover, eliminating API rate-limit bottlenecks for continuous summary generation.
- Built an interactive Streamlit dashboard featuring custom Neutral/Mixed sentiment thresholding with dynamic confidence range sliders, real-time latency metrics, and batch CSV processing.

Ollanium – Prompt-Driven Web Intelligence Pipeline

Ollama (Llama 3.1), Selenium, BeautifulSoup4, Streamlit

- Built an air-gapped web intelligence pipeline merging Selenium browser automation with local Ollama inference, replacing fragile site-specific CSS selectors with natural-language extraction prompts.
- Integrated a BeautifulSoup4 sanitization pipeline to strip structural boilerplate, minimizing prompt noise and optimizing context-window token density for Llama 3.1.
- Developed a Streamlit orchestration UI to submit target URLs, monitor real-time scraping execution logs, and instantly return structured data extracted via conversational English instructions.

Skills

AI & LLM Engineering: LangGraph, Agentic Orchestration, Prompt Engineering, LLMs (Gemini, Llama 3.3, GPT), Groq API, Scikit-Learn (LinearSVC), NLTK, Selenium, BeautifulSoup4, Streamlit

Languages & Databases: Python, TypeScript, JavaScript, C++, SQL, MongoDB

Tools & Platforms: Git, GitHub, Postman, VS Code, Cursor, Antigravity

Achievements

- Solved **400+** Data Structures & Algorithms questions on LeetCode and Codeforces, demonstrating strong technical problem-solving fundamentals.
- Core Team Member in **DesignAndCode** committee.